Final Review

Number Theory

Problem 1

In each of the following, find a value of x that satisfies the given congruence, or argue that no such x exists. Show how you found x.

a. $4(x-3) \equiv 8x - 3 \pmod{41}$

b. $7(x+5) \equiv 2x \pmod{14}$

Solution

a. $4(x-3) \equiv 8x - 3 \pmod{41}$

$4x - 12 \equiv 8x - 3 \pmod{41}$

$4x \equiv 8x + 9 \pmod{41}$

$-4x \equiv 9 \pmod{41}$

$4x \equiv -9 \pmod{41}$

$4x \equiv 32 \pmod{41}$

Here, we can conclude that $x = 8$ is a possible answer by inspection. Or, we can find the multiplicative inverse.

Now we know that 41 is prime. Hence, 4 has a multiplicative inverse. We use the tabular Euclidean Algorithm to find the gcd of $4, 41$ to find the multiplicative inverse of $4 \pmod{41}$.

a	b	s	t	g
4	41	-10	1	1
1	4	1	0	1
0	1	0	1	1

Thus, -10 is the multiplicative inverse of $4 \pmod{41}$.

$$x \equiv -320 \pmod{41}$$

Hence, $x = -320$ is a possible answer.

b. $7(x+5) \equiv 2x \pmod{14}$

$7x + 35 \equiv 2x \pmod{14}$

$5x \equiv -35 \pmod{14}$

$5x \equiv 7 \pmod{14}$

We now use the tabular Euclidean Algorithm to find the gcd of $5, 14$ to find the multiplicative inverse of $5 \pmod{14}$.

a	b	s	t	g
5	14	3	-1	1
4	5	-1	1	1
1	4	1	0	1
0	1	0	1	1

Thus, 3 is the multiplicative inverse of $5 \pmod{14}$.

$$x \equiv 21 \pmod{14}$$

$$x \equiv 7 \pmod{14}$$

Hence, $x = 7$ is a possible answer.

Problem 2

The value of the Euler ϕ function at the positive integer n is defined to be the number of positive integers less than or equal to n that are relatively prime to n.

a. Find the following values:

(i) $\phi(8)$

(ii) $\phi(7)$

(iii) $\phi(15)$

b. Prove that n is composite iff $\phi(n) < n - 1$.

Solution

a. Find the following values:

(i) $\phi(8) = 4$, as 1, 3,5, and 7 are smaller positive integers than 8 that are also relatively prime to 8.

(ii) $\phi(7) = 6$ because primes are relatively prime to all positive integers smaller than them.

(iii) $\phi(15) = \phi(3) \cdot \phi(5) = (3 - 1) \cdot (5 - 1) = 8$

b. To prove iff, we must prove both directions.

$$n \text{ composite} \implies \phi(n) < n - 1$$

n composite means there is some $p \in \mathbb{Z}$ $1 < p < n$ such that $p \mid n$. As such, $\phi(n)$ is at most $n - 2$, which means that $\phi(n) < n - 1$

$$n \text{ not composite} \implies \phi(n) \not< n - 1$$

If n is not composite, then n is must be either 1 or a prime number.

If $n = 1$, then $\phi(n) = \phi(1) = 1 = n$, which means that $\phi(n) \not< n - 1$

Similarly, if n is a prime integer, then $\phi(n) = n - 1$, which means that $\phi(n) \not< n - 1$. In either case, the condition holds.

Conclusion: Since both directions hold, we have proven that n is composite iff $\phi(n) < n - 1$.

Problem 3

Suppose $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$. Let $n \in \mathbb{Z}^+$. Use the definition of mod to prove the following statements:

a. $ac \equiv bd \pmod{m}$.

b. $a^n \equiv b^n \pmod{m}$.

Hint: Try induction on n.

Solution

a. **Claim:** $ac \equiv bd \pmod{m}$.

By the definition of mod, $\exists\ k, l \in \mathbb{Z}$ such that

$$a = b + km$$
$$c = d + lm$$

Multiplying these equations gives:

$$ac = (b + km)(d + lm)$$
$$ac = bd + blm + kmd + klm^2$$
$$ac = bd + m(bl + kd + klm)$$

Notice that the second term on the RHS is divisible by m. Therefore, we may rewrite the equation above in the form

$$ac \equiv bd \pmod{m}.$$

b. **Claim:** $\forall n \in \mathbb{Z}^+, a^n \equiv b^n \pmod{m}$.

We prove this statement by induction on the exponent, n.

Base Case: $n = 1$. It is given that $a \equiv b \pmod{m}$, and since $x = x^1$ for all integers, we can conclude that $a^1 \equiv b^1 \pmod{n}$.

Inductive Step: Suppose $a^k \equiv b^k \pmod{m}$ for some $k \in \mathbb{Z}^+$. We must now show that $a^{k+1} \equiv b^{k+1} \pmod{m}$ holds. Note that we may rewrite a^{k+1} as $a \cdot a^k$. Likewise, we may rewrite b^{k+1} as $b \cdot b^k$. Using what we proved in part a (and substituting a^k for c and b^k for b, we know that since $a \equiv b \pmod m$ and $a^k \equiv b^k \pmod m$, we can conclude that

$$a \cdot a^k \equiv b \cdot b^k \pmod m.$$

It follows that

$$a^{n+1} \equiv b^{n+1} \pmod m$$

as needed.

We have shown that the base case and the inductive step hold, so the claim holds for all $n \in \mathbb{Z}^+$.

Counting

Problem 1

Consider a finite set S. Count the number of subset pairs for which the two pairs do not overlap, justifying your answer. In other words, subsets $s_1, s_2 \in S$ form the valid pair $\{s_1, s_2\}$ iff $s_1 \cap s_2 = \varnothing$.

Note: The order within the pair does not matter.

Solution 1:

Consider two empty sets a and b that we can fill with elements to create two non-overlapping subsets.

For all $e \in S$, we can choose to put e either in a, in b, or in neither. This guarantees that there is no overlap between a and b, and counts all possible valid subset pairs (a, b). Since there are three options for every element $e \in S$, there are 3^n ways of doing this. Notice that (a, b) and (b, a) are counted as distinct pairs for $a \neq b$, so we are double counting.

Consider then the edge case where $a = b$, which only occurs when $a = b = \varnothing$ since we require that $a \cap b = \varnothing$. This pair is not double counted. Hence, we can add one to double count the pair $(\varnothing, \varnothing)$, and then divide the total by $2! = 2$ to undo the double counting on all pairs. Hence, we conclude that the number of non-overlapping pairs is

$$\frac{3^n + 1}{2}.$$

Solution 2:
Consider a subset $a \in S$ of size $|a| = k$, where $0 \leq k \leq n$. To pair this subset up with another subset such that they don't overlap, the other subset must not contain any element $e \in a$. Hence, we can pair a with any subset $b \subseteq \overline{a}$ such that $\forall e \in b, e \notin a$.

There are $\binom{n}{k}$ ways to pick a subset a of size k. Since $b \in \mathcal{P}(\overline{a})$, we find that there are 2^{n-k} ways to choose a subset b with to pair with a. Since $0 \leq k \leq n$, we conclude that there are

$$\sum_{k=0}^{n} 2^{n-k} \binom{n}{k}$$

ways to choose a valid pair a and b. Notice that the pairs (a, b) and (b, a) for $a \neq b$ are counted as distinct, so we are double counting. For the edge case where $a = b$, which only occurs when $a = b = \varnothing$ since we require that $a \cap b = \varnothing$, the pair $(\varnothing, \varnothing)$ is not double counted. Hence, we can add one to double count $(\varnothing, \varnothing)$, and then divide the total by $2! = 2$ to undo all double counting. We conclude that the number of ways in which we can pick pairs of non-overlapping subsets of S is given by

$$\frac{1}{2}\left[1 + \sum_{k=0}^{n} 2^{n-k}\binom{n}{k}\right].$$

Problem 2

Consider n distinct objects. How many distinct sequences are there of these n non-repeating items (any length)?

Solution

Consider a subset s of these n items of size $|s| = k$, where $0 \leq k \leq n$. There are $\binom{n}{k}$ ways to choose such a subset. Once this subset has been chosen, there are $k!$ ways to arrange them into a sequence. Hence, the number of distinct sequences of any length of these n non-repeating items is

$$\sum_{k=0}^{n} k! \binom{n}{k}$$

Problem 3

How many different ways are there to list the numbers $1, 2, \ldots, 2n$ such that the even numbers appear in increasing order and the odd numbers appear in decreasing order?

Solution

There are n even numbers and n odd numbers. Considering these groups separately, there is only one way to order them in increasing and decreasing order, respectively.

Consider $2n$ spots in which we can put these numbers. There are $\binom{2n}{n}$ ways to choose n of those spots to put the n even numbers increasing order. The remaining n spots are then filled with the n odd numbers in decreasing order. Hence, we conclude that there are $\binom{2n}{n}$ ways to list the numbers $1, 2, \ldots, 2n$ such that the even numbers appear in increasing order and the odd numbers appear in decreasing order

Probability

Problem 1

Consider a line of n aliens at a spaceport looking to board a rocket containing n free seats. Each of them has a ticket, and the i^{th} alien is assigned to the i^{th} seat.

The first alien in line enters the rocket, and instead of looking at its ticket for its assigned seat, sits uniformly at random. Every following alien that enters chooses its assigned seat if it is available, otherwise chooses one of the remaining seats uniformly at random. What is the probability that the n^{th} alien sits down in its assigned seat?

Solution

Every alien either sits in its assigned seat or picks one of the remaining seats uniformly at random. Note that the n^{th} alien sits down in whatever seat remains. Since the first alien picks uniformly at random regardless of circumstance, there is always at least one alien that picks a seat at random.

Let us consider the i^{th} alien having to choose a seat out of the remaining seats uniformly at random, where $1 \leq i < n$. There are $n - i + 1$ remaining seats. There are now three cases to consider.

1. If the i^{th} alien chooses any seat s where $2 \leq s \leq n - 1$, then it displaces another alien in line. Thus, we return to these three cases recursively.
2. If the i^{th} alien chooses the first seat with probability $\frac{1}{n-i+1}$, then the first alien and the i^{th} alien have essentially switched seats. Hence, no remaining alien in line is displaced, the shuffling terminates, and the n^{th} alien sits down in its assigned n^{th} seat.
3. If the i^{th} alien chooses the n^{th} seat with probability $\frac{1}{n-i+1}$, then the n^{th} alien necessarily sits in the 1^{st} seat.

Note here that we eventually reach a terminating condition where the n^{th} alien is assigned a seat. For every i^{th} alien considered, we find two equally likely cases: the n^{th} alien either sits down in the n^{th} seat with probability $\frac{1}{n-i+1}$ or sits down in the 1^{st} seat with probability $\frac{1}{n-i+1}$.

Hence, we conclude that the n^{th} alien sits down the n^{th} seat with probability $\frac{1}{2}$.

Problem 2

Of 100 coins on a table, 99 of them are fair and 1 of them has H on both sides.

a. You choose a coin uniformly at random, and flip it. Let X be a random variable where $X = 1$ if you get heads, and $X = 0$ if you get tails. What is $\mathbb{E}[X]$?

b. You reset the coins and choose a coin again uniformly at random, flip it, record the result, and put the coin back. You repeat this experiment 9 more times (for a total of 10). Let X be a random variable denoting the total number of heads you get. What is $\mathbb{E}[X]$?

c. You reset the coins and choose a coin again uniformly at random, but flip this *same coin* 10 times. If you get heads every time, what is the probability you will also get heads the 11th time?

Solution

a. The probability we get heads is $Pr(H|\text{fair coin})Pr(\text{fair coin})$ $+ Pr(H|\text{biased coin})Pr(\text{biased coin})$.

- $Pr(H|\text{fair coin}) = \frac{1}{2}$
- $Pr(\text{fair coin}) = \frac{99}{100}$
- $Pr(H|\text{biased coin}) = 1$
- $Pr(\text{biased coin}) = \frac{1}{100}$

The total probability is 0.505.

b. Let X_i be a random variable that is 1 if the i-th flip is heads, and 0 otherwise. $X = \sum_{i=1}^{10} X_i$.

By the linearity of expectation, $\mathbb{E}[X] = \sum_{i=1}^{10} \mathbb{E}[X_i]$. From part a, each $\mathbb{E}[X_i] = 0.505$. So, $\mathbb{E}[X] = 5.05$.

c. We want to find Pr(H 11th time|10 H). This quantity can be expanded into two cases: whether the coin we picked is a fair coin or the biased coin. We also have to factor in the probability of the coin being fair or unfair based on the fact that we've just gotten 10 heads in a row. So, we are looking for:

$$\Pr(\text{H 11th time}|\text{fair coin})\,\Pr(\text{fair coin}|\text{10 H}) +$$

$$\Pr(\text{H 11th time}|\text{unfair coin})\,\Pr(\text{unfair coin}|\text{10 H})$$

Using Bayes' Theorem, we can calculate each component:

$$= .5 \times \frac{\Pr(10 \text{ H}|\text{fair coin}) \Pr(\text{fair coin})}{\Pr(10 \text{ H})} + 1 \times \frac{P(10 \text{ H}|\text{biased coin}) P(\text{biased coin})}{\Pr(10 \text{ H})}$$
$$= \frac{.5 \times .5^{10} \times .99 + 1 \times 1 * .01}{\Pr(10 \text{ H}|\text{fair coin}) \Pr(\text{fair coin}) + \Pr(10 \text{ H}|\text{biased coin}) \Pr(\text{biased coin})}$$
$$= \frac{.5^{11} \times .99 + .01}{.5^{10} \times .99 + 1 * .01} = \frac{.5^{11} \times .99 + .01}{.5^{10} \times .99 + .01}$$
$$= 0.956$$

Problem 3

Flip a coin repeatedly and consider the sequence of heads or tails you land. Stop flipping as soon as you get an HHT or an HTT. Which one is more likely to cause you to stop?

Solution

Let A be the event where HHT occurs before HTT occurs. By the law of total probability, we have

$$\begin{aligned} \Pr(A) &= \Pr(H) \Pr(A|H) + \Pr(T) \Pr(A|T) \\ &= \frac{1}{2} \Pr(A|H) + \frac{1}{2} \Pr(A) \end{aligned}$$

$$\Rightarrow \Pr(A) = \Pr(A|H).$$

Note here that $\Pr(A) = \Pr(A|T)$ since flipping tails is not the start of either sequence. We have $\Pr(A) = \Pr(A|H)$, since the probability that A occurs depends solely on what follows the first heads you flip. Continuing with the sequence, we have

$$\begin{aligned} \Pr(A) &= \Pr(H) \Pr(A|HH) + \Pr(T) \Pr(A|HT) \\ &= \frac{1}{2} \Pr(A|HH) + \frac{1}{2} \Pr(A|HT). \end{aligned}$$

Now notice that $P(A|HH) = 1$, since, regardless of what we flip next, HHT occurs before HTT; if we flip *HH*, then even if we flip heads again, we are always one flip closer to HHT than HTT. Thus,

$$\Pr(A) = \frac{1}{2} + \frac{1}{2}\Pr(A|HT).$$

Let us now consider $\Pr(A|HT)$.

$$\begin{aligned}\Pr(A|HT) &= \Pr(H)\Pr(A|HTH) + \Pr(T)\Pr(A|HTT)\\ &= \frac{1}{2}\Pr(A|H) + \frac{1}{2}\cdot 0\end{aligned}$$

$$\Rightarrow \Pr(A|HT) = \frac{1}{2}\Pr(A).$$

Hence, we conclude that

$$\Pr(A) = \frac{1}{2} + \frac{1}{2}\left(\frac{1}{2}\Pr(A)\right) \quad \Rightarrow \quad \Pr(A) = \frac{2}{3}.$$

Hence, since $Pr(A) > \frac{1}{2}$, HHT is more likely to occur before HTT.

Graph Theory

Problem 1

Show that, in any graph G, if there is a vertex v of odd degree, then there is a path from v to some other vertex of odd degree.

Solution

Recall from lecture that, because vertices share edges, the sum of the degrees in a graph must be equal to $2|E|$. Hence, we know that if there is one vertex of odd degree, there must be at least one other. (The property also holds for the connected component containing the vertex of odd degree.)

We can construct a path starting at one of the odd-degree vertices v in the following manner: follow any edge from v. There are now an even number of unexplored edges incident to v. Continue following unexplored edges in this way. For each vertex we visit, there are two cases.

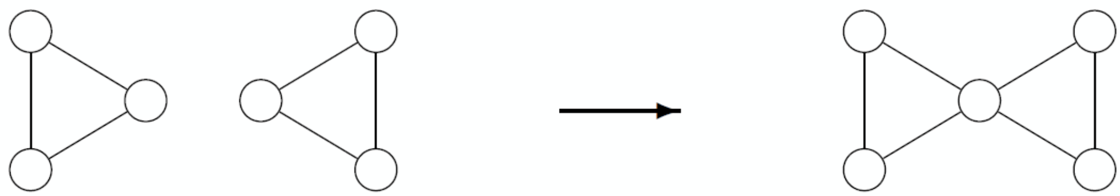
1. The vertex is an odd-degree vertex, as needed, and we are done.
2. The vertex is an even-degree vertex (or our original vertex). We use two edges to enter and leave the vertex, meaning this leaves this vertex with an even number of unexplored edges.

Because we cannot enter an even-degree vertex and not have another edge upon which to leave, we must eventually reach a vertex of odd degree.

Thus, there is always a path between odd-degree vertices in a simple graph if it has an odd-degree vertex.

Problem 2

Prove that by joining together any $n \geq 2$ connected graphs that have connected Eulerian tours the resulting graph will also have a Eulerian tour. In this problem, we define joining as merging any two vertices together into a single new vertex, as depicted in the image below:



Solution

We can also prove this property by using induction.

Proof. **Proposition:** Let $P(n)$ be the predicate that joining any n connected graphs each with a connected Eulerian tour results in a new graph with a Eulerian tour, for $n \geq 2$.

Base Case: Consider the case $n = 2$.

Let $n = 2$. Consider two graphs G_1 and G_2 that have a Eulerian tour. Recall from the lecture on July 23^{rd} that they both have a Eulerian tour iff their vertices all have an even degree.

Without loss of generality, let us assume that we merge the two at vertex v_1 in G_1 and v_2 in G_2. Since v_1 and v_2 have an even degree, they have $2e_1$ and $2e_2$ outgoing edges where $e_1, e_2 \in \mathbb{N}$, respectively. When we merge the two graphs at a vertex, we do not lose any of those edges. The resulting merged vertex has a total of $2e_1 + 2e_2 = 2(e_1 + e_2)$ edges, which is even. Thus, the resulting graph G' from the merge has vertices that all have an even degree, so it must have a Eulerian tour, and $P(2)$ holds.

Inductive Hypothesis: Assume $P(k)$ is true for $n = k$.

Inductive Step: We want to show $P(k + 1)$ is true, ie. that a graph resulting from connecting $k+1$ connected graphs with a Eulerian tour has a Eulerian tour.

Consider $k+1$ connected graphs with a Eulerian tour. Take k of these $k+1$ graphs and connect them. By the inductive hypothesis, the result is a graph G_k with a Eulerian tour. Hence, G_k and the $k+1^{th}$ graph both have vertices with an even degree. Let us merge the two at a vertex to form G_{k+1}. As explained in the base case, the merged vertex also has an even degree. Hence, the resulting graph G_{k+1} has vertices with an even degree, and we conclude that G_{k+1} has a Eulerian tour and that $P(k+1)$ holds.

Conclusion: We have shown that $P(2)$ is true, and that if $P(k)$ is true then so is $P(k+1)$. Hence, $P(n)$ is true for all $n \geq 2$ by the principle of induction.

$\square$

Problem 3

Pluto is visiting some friends around the solar system. There exists spaceflights between friends in Pluto's group. In fact, there are enough for Pluto's group to be 'connected'; it is possible to go from any friend to any other friend through some sequence of spaceflights.

A fanatic of rockets, Pluto looks to begin the trip from its home and take every possible spaceflight. Consider the connected graph $G = (V, E)$, where V is the set of Pluto's friend group, and E is the set of spaceflights between them (the graph is undirected; we assume that if a spaceflight exists between two friends, it exists in both directions).

To keep things cheap, Pluto will attempt its plan to take every spaceflight if and only if there exists a Eulerian tour through them. You will now work to prove that Pluto will attempt the trip if and only if each of its friends have an even number of spaceflights connecting them to other friends in the group.

a. Prove that if G has a Eulerian tour, then every vertex $v \in V$ has even degree. (This result was given in lecture, but try it for yourself!)

b. Prove, by induction on the number of edges, that if every vertex $v \in V$ has even degree, then G has an Eulerian tour.

   Note: You should use **build-down** induction!

Solution

a. *Proof.* Let G be a graph that has a Eulerian tour. Without loss of generality, suppose for the sake of contradiction that some vertex $v \in G$ does not have even degree. Let R be the Eulerian tour that starts and ends at v.

   Since R starts and ends on v, there must be an initial edge by which it leaves v and a final edge by which it ends at v. It may visit v again any time during its cycle, but by the same logic as above, must enter and exit v by a pair of distinct, non-traversed edges. However, this implies that v must have an even number of edges, contradicting our assumption that v has odd degree.

   Thus no such v exists, and every vertex in V has even degree. □

b. We prove this by induction on the number of edges.

   Proof. **Base Case:** Consider the graph with zero edges. Since this graph is connected, it must be a single vertex. This vertex has zero incident edges, which means it has even degree. This graph has a Eulerian cycle consisting just of the one vertex. Thus our base case holds.

Inductive Hypothesis: Suppose all simple connected graphs with fewer than or equal to n edges and with vertices all of even degree have Eulerian circuits.

Inductive Step: Let $G = (V, E)$ be a simple connected graph with $n + 1$ edges, where every vertex has even degree.

Fix some starting vertex $v \in V$. From v, continually traverse edges in E, being sure not to traverse any edge more than once, until you arrive back at v. Since every vertex has even degree, it is impossible to reach a dead end before you revisit v: for every vertex there is a pair of incident edges by which to enter and exit. Call this cycle C.

Now, form a new graph $G' = (V, E')$ by removing all the edges traversed in C. Note that G' consists of one or more connected components; let us call these subgraphs $G_1, G_2, \dots, G_k$. Note that, for every vertex $u \in V$, C visited u the same number of times it left u. Thus, in obtaining E' from E, every node has its degree reduced by an even number; every node must, then, still have even degree. Note further that, since we've removed the edges in C, each of $G_1, G_2, \dots, G_k$ has fewer than $n + 1$ edges, i.e. at most n edges.

Hence, by the inductive hypothesis, each G_i has a Eulerian cycle. The Eulerian cycles on G_i together with the original cycle C cover every edge in E exactly once.

Consider concatenating C and the Eulerian cycles of the G_i in the following manner: continue along C until you reach a vertex u that is in one of the subgraphs $G_1, G_2, \dots, G_k$, say G_u. At this point, follow the Eulerian cycle of G_u, which will end back at u. Continue along C. Repeat this process all along C, but do not follow any Eulerian cycle more than once. Since G is connected, every subgraph G_i will have a node reached by C, and thus every Eulerian cycle will at some point be concatenated. We have already claimed that C together with these Eulerian cycles cover every edge in E. Since C ends at starting vertex v, this concatenation is a Eulerian cycle for G.

Conclusion: Thus, since the property holds for zero edges, and since if the property holds for n edges it holds for $n + 1$ edges, then any simple graph G with $n \geq 0$ edges such that every vertex has an even degree has a Eulerian tour. $\square$

Note: Keep in mind that *building up* is not enough for this induction proof, because it would only prove the claim for specific graphs rather than all graphs with $n + 1$ edges.